

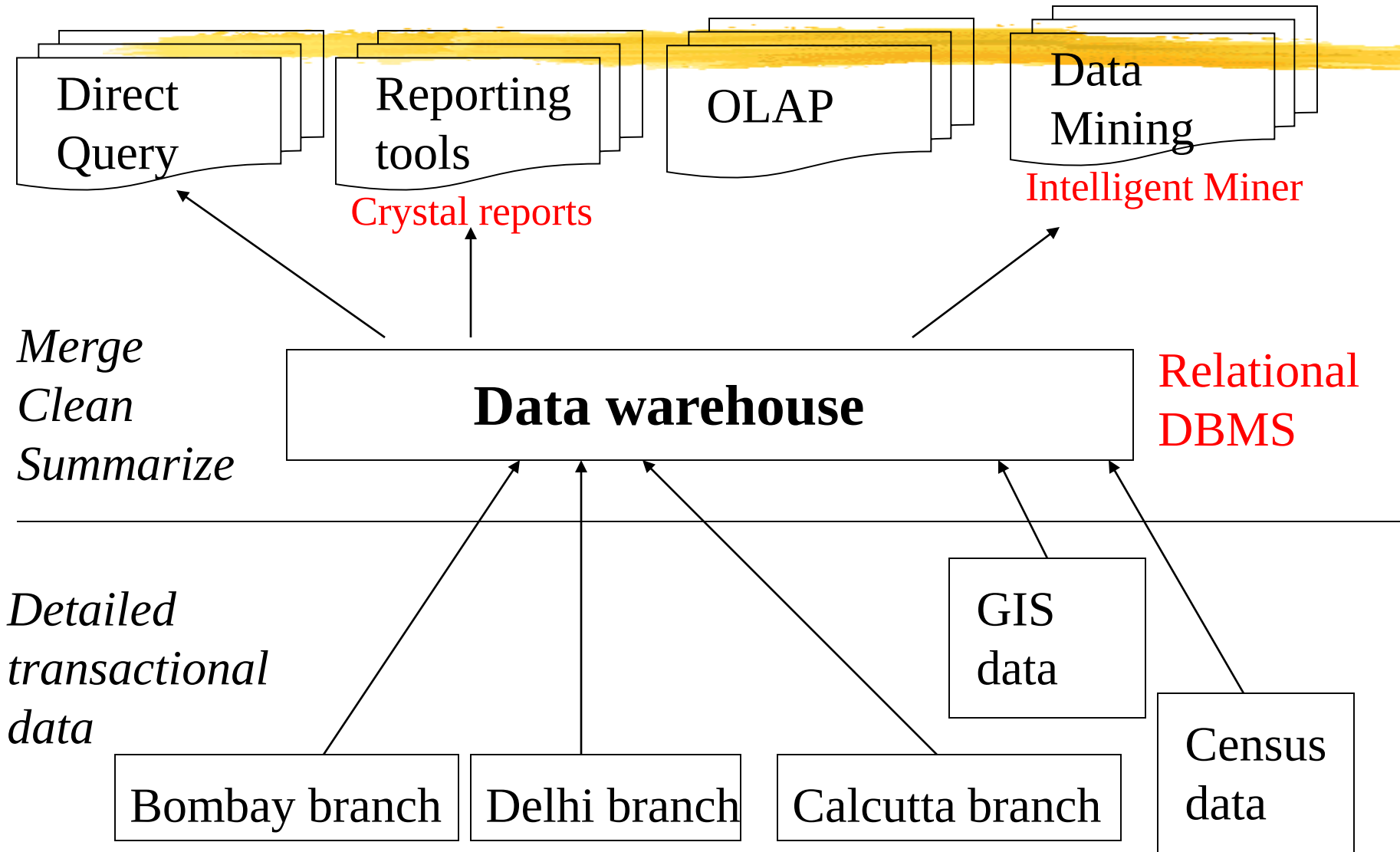


Data warehousing

What is Data Warehouse?

- ⌘ Defined in many different ways, but not rigorously.
 - ☒ A decision support database that is maintained **separately** from the organization's operational database
 - ☒ Support **information processing** by providing a solid platform of consolidated, historical data for analysis.
- ⌘ "A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process."—W. H. Inmon
- ⌘ Data warehousing:
 - ☒ The process of constructing and using data warehouses

Decision support tools



Operational data

Decision support systems



- ⌘ Data base systems two types
 - ☒ Transaction processing systems: OLTP
 - ☒ Decision support systems: warehouses, OLAP, Mining
- ⌘ Historical data from OLTP systems form decision support systems
- ⌘ Goal: Learn from experiences in historical data
- ⌘ OLTP: many short, update intensive commands
- ⌘ Decision support: fewer, larger queries.

Data Warehouse—Subject-Oriented

- ⌘ Organized around major subjects, such as **customer, product, sales**.
- ⌘ Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing.
- ⌘ Provide **a simple and concise** view around particular subject issues by **excluding data that are not useful in the decision support process**.

Data Warehouse—Integrated



- ⌘ Constructed by integrating multiple, heterogeneous data sources
 - ☒ relational databases, flat files, on-line transaction records
- ⌘ Data cleaning and data integration techniques are applied.
 - ☒ Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
 - ☒ E.g., Hotel price: currency, tax, breakfast covered, etc.
 - ☒ When data is moved to the warehouse, it is converted.

Data Warehouse—Non-Volatile

- ⌘ A **physically separate store** of data transformed from the operational environment.
- ⌘ Operational **update of data does not occur** in the data warehouse environment.
 - ☒ Does not require transaction processing, recovery, and concurrency control mechanisms
 - ☒ Requires only two operations in data accessing:
 - ☒ *initial loading of data* and *access of data*.

Types of Data warehouse

⌘ Enterprise Data Warehouse (EDW)

EDW is a centralized warehouse that enables decision support services across the enterprise. It is used when a unified approach for representing and organizing data is needed. A major advantage of EDW is that it classifies data based on subject and provides access to those divisions, enabling better data governance and management.

⌘ Operational Data Store:

Operational Data Store, which is also called ODS, are nothing but data store required when neither Data warehouse nor OLTP systems support organizations reporting needs. In ODS, Data warehouse is refreshed in real time. Hence, it is widely preferred for routine activities like storing records of the Employees.

⌘ Data Mart:

A data mart is a subset of the data warehouse. It specially designed for a particular line of small business or individual modules such as sales, finance, sales or finance. In an independent data mart, data can collect directly from sources.

Data Warehouse vs. Heterogeneous DBMS

⌘ Traditional heterogeneous DB integration:

- ☒ Build **wrappers/mediators** on top of heterogeneous databases
- ☒ **Query driven** approach
 - ☒ When a query is posed to a client site, a meta-dictionary is used to translate the query into queries appropriate for individual heterogeneous sites involved, and the results are integrated into a global answer set
 - ☒ Complex information filtering, compete for resources

⌘ Data warehouse: **update-driven**, high performance

- ☒ Information from heterogeneous sources is integrated in advance and stored in warehouses for direct query and analysis

Data Warehouse vs. Operational DBMS

⌘ OLTP (on-line transaction processing)

- ☒ Major task of traditional relational DBMS
- ☒ Day-to-day operations: purchasing, inventory, banking, manufacturing, payroll, registration, accounting, etc.

⌘ OLAP (on-line analytical processing)

- ☒ Major task of data warehouse system
- ☒ Data analysis and decision making

⌘ Distinct features (OLTP vs. OLAP):

- ☒ User and system orientation: customer vs. market
- ☒ Data contents: current, detailed vs. historical, consolidated
- ☒ Database design: ER + application vs. star + subject
- ☒ View: current, local vs. evolutionary, integrated
- ☒ Access patterns: update vs. read-only but complex queries

OLTP vs. OLAP

	OLTP	OLAP
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date detailed, flat relational isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions
#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response

Why Separate Data Warehouse?

⌘ High performance for both systems

- ☒ DBMS— tuned for OLTP: access methods, indexing, concurrency control, recovery
- ☒ Warehouse—tuned for OLAP: complex OLAP queries, multidimensional view, consolidation.

⌘ Different functions and different data:

- ☒ missing data: Decision support requires historical data which operational DBs do not typically maintain
- ☒ data consolidation: DS requires consolidation (aggregation, summarization) of data from heterogeneous sources
- ☒ data quality: different sources typically use inconsistent data representations, codes and formats which have to be reconciled

Conceptual Modeling of Data Warehouses

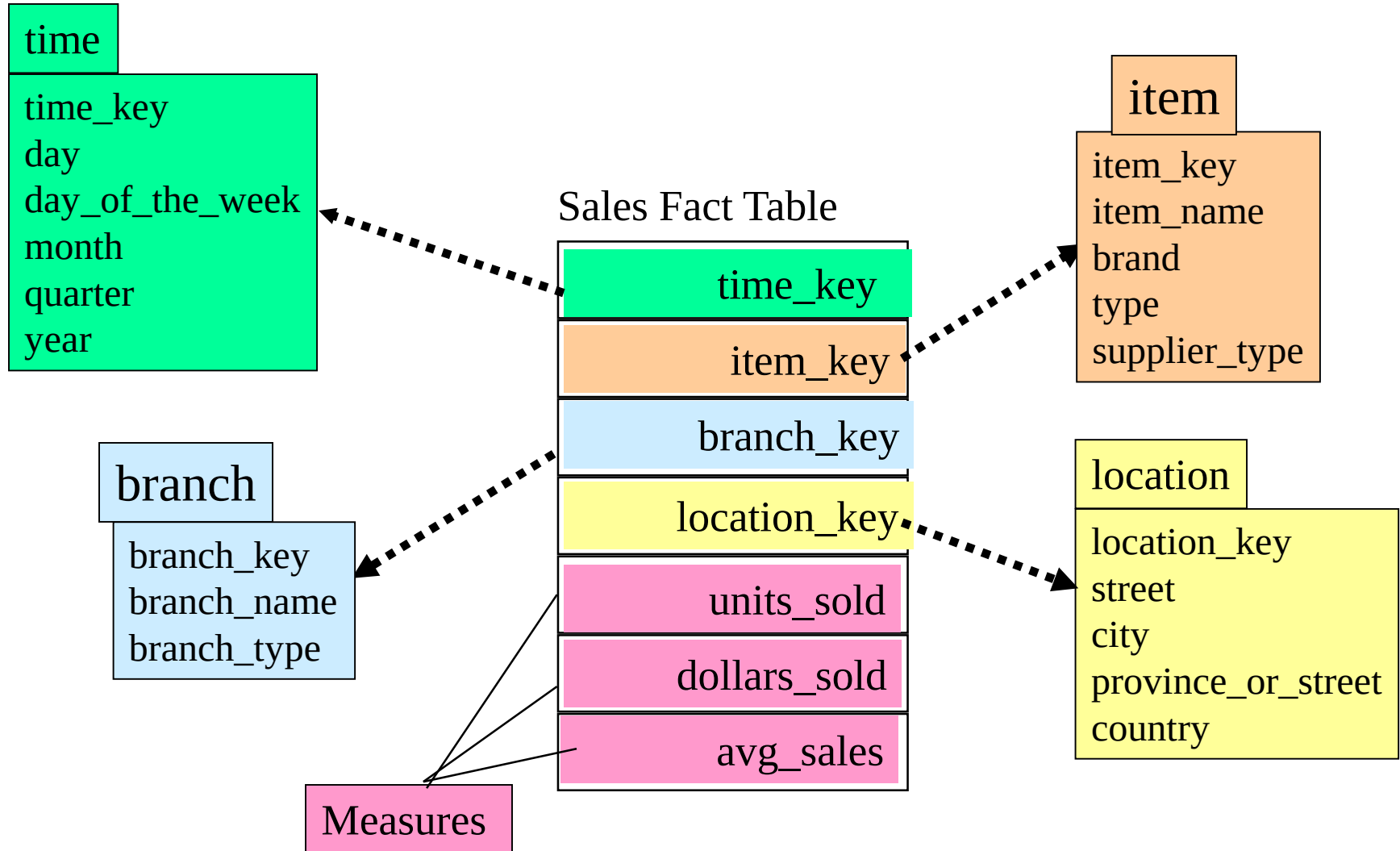
⌘ Modeling data warehouses: dimensions & measures

☒ Star schema: A fact table in the middle connected to a set of dimension tables

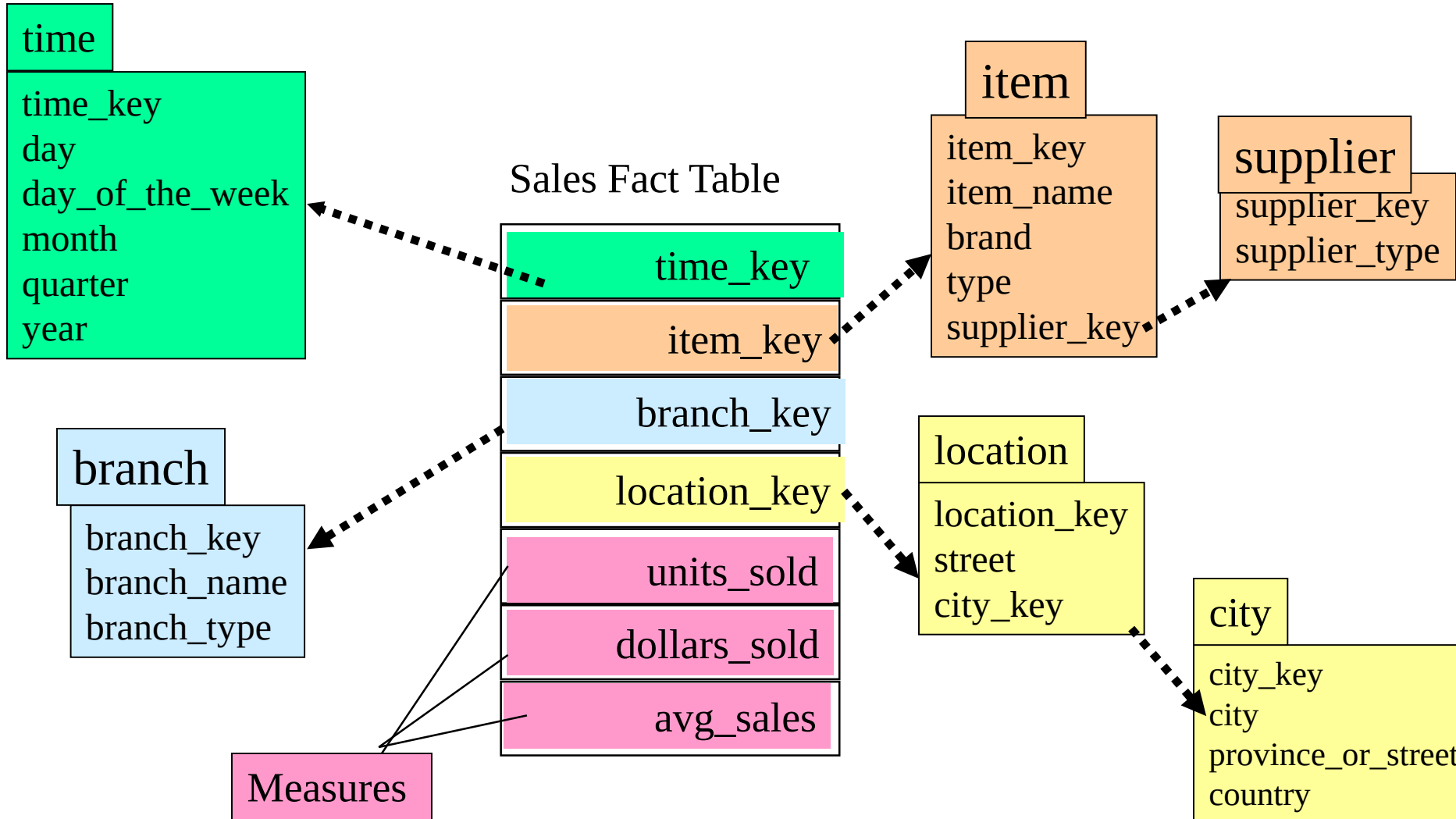
☒ Snowflake schema: A refinement of star schema where some dimensional hierarchy is **normalized** into a set of smaller dimension tables, forming a shape similar to snowflake

☒ Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called **galaxy schema** or fact constellation

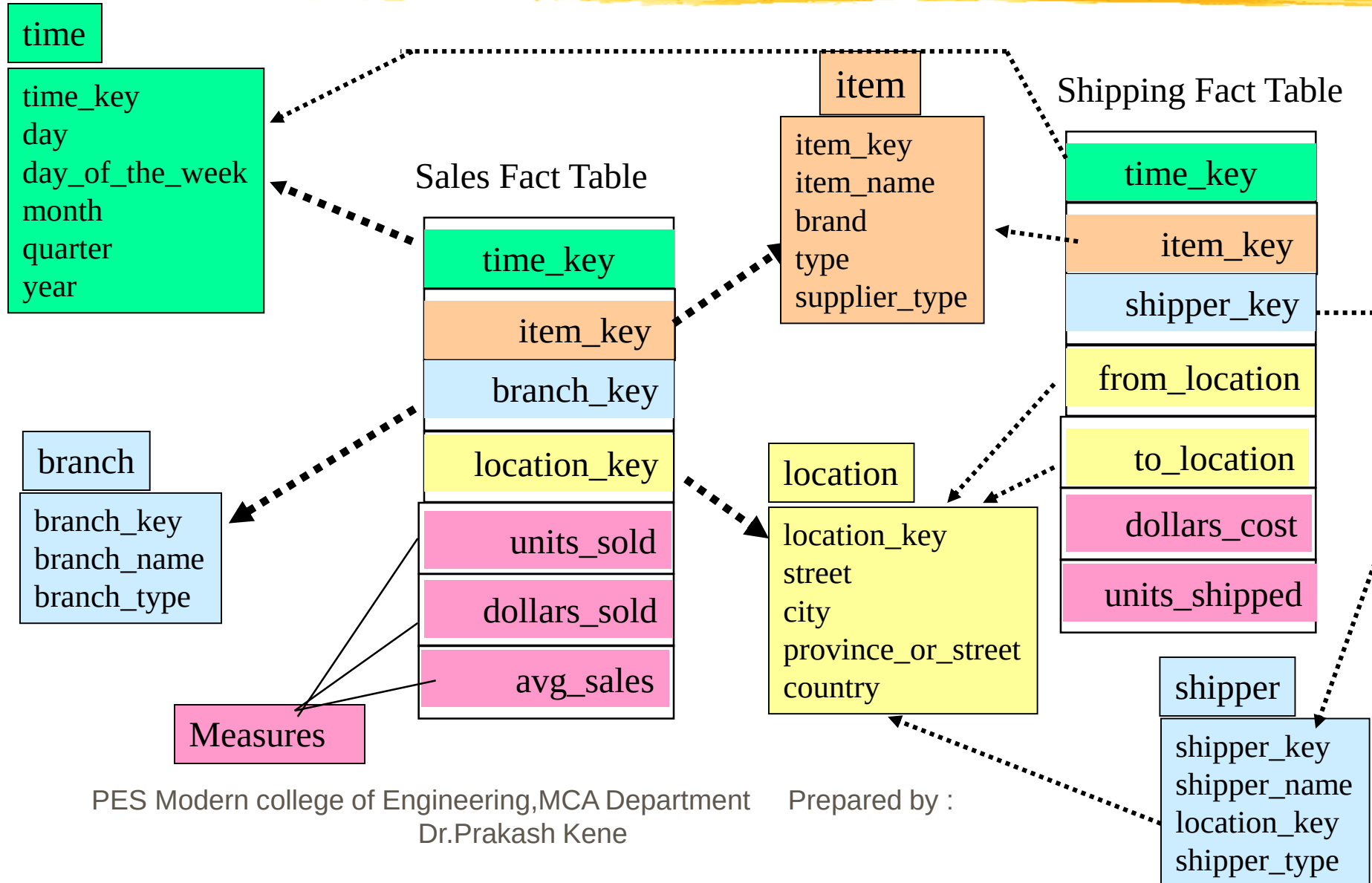
Example of Star Schema



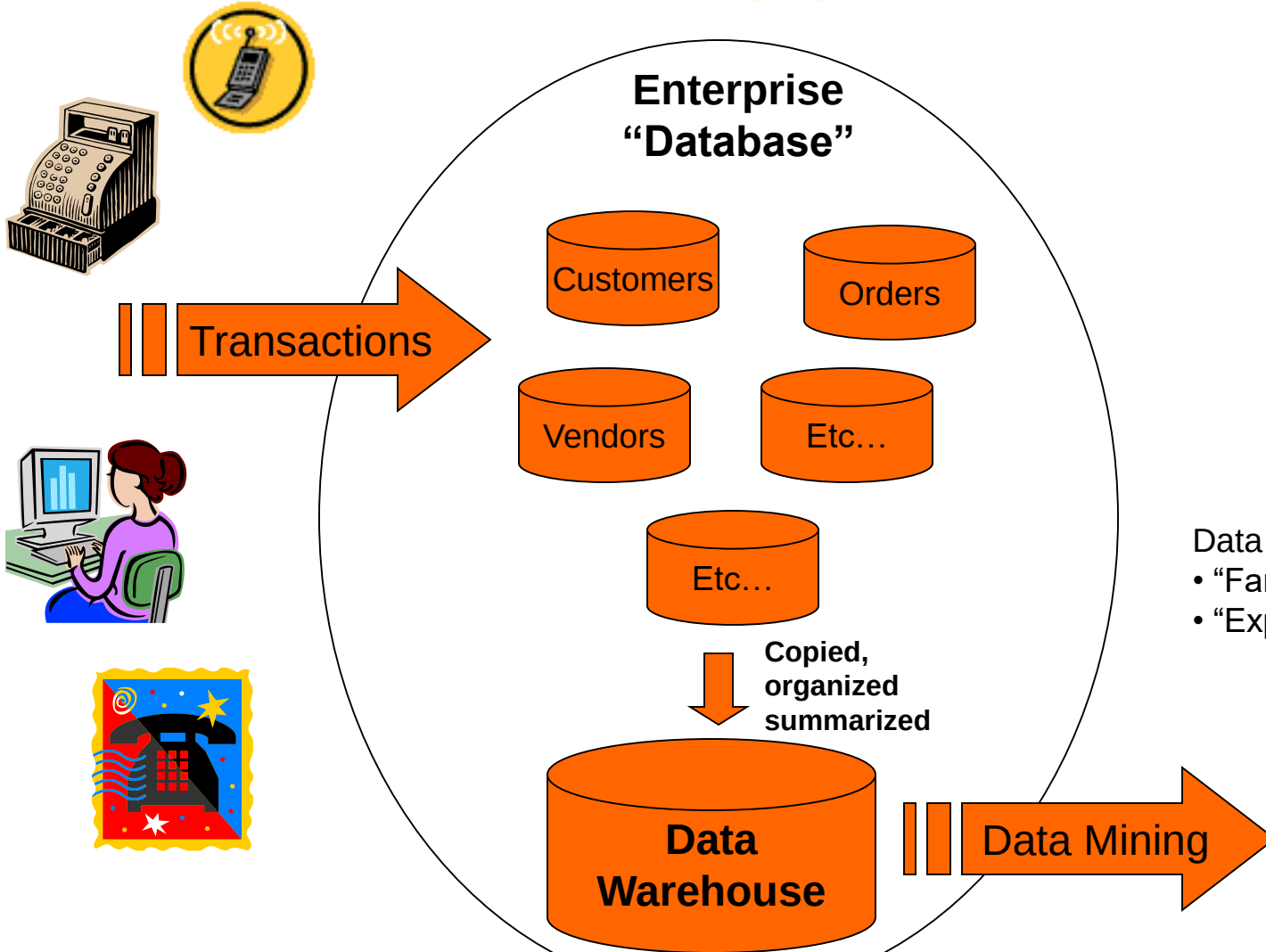
Example of Snowflake Schema



Example of Fact Constellation



Data Warehouse



Data Miners:

- "Farmers" – they know
- "Explorers" - unpredictable

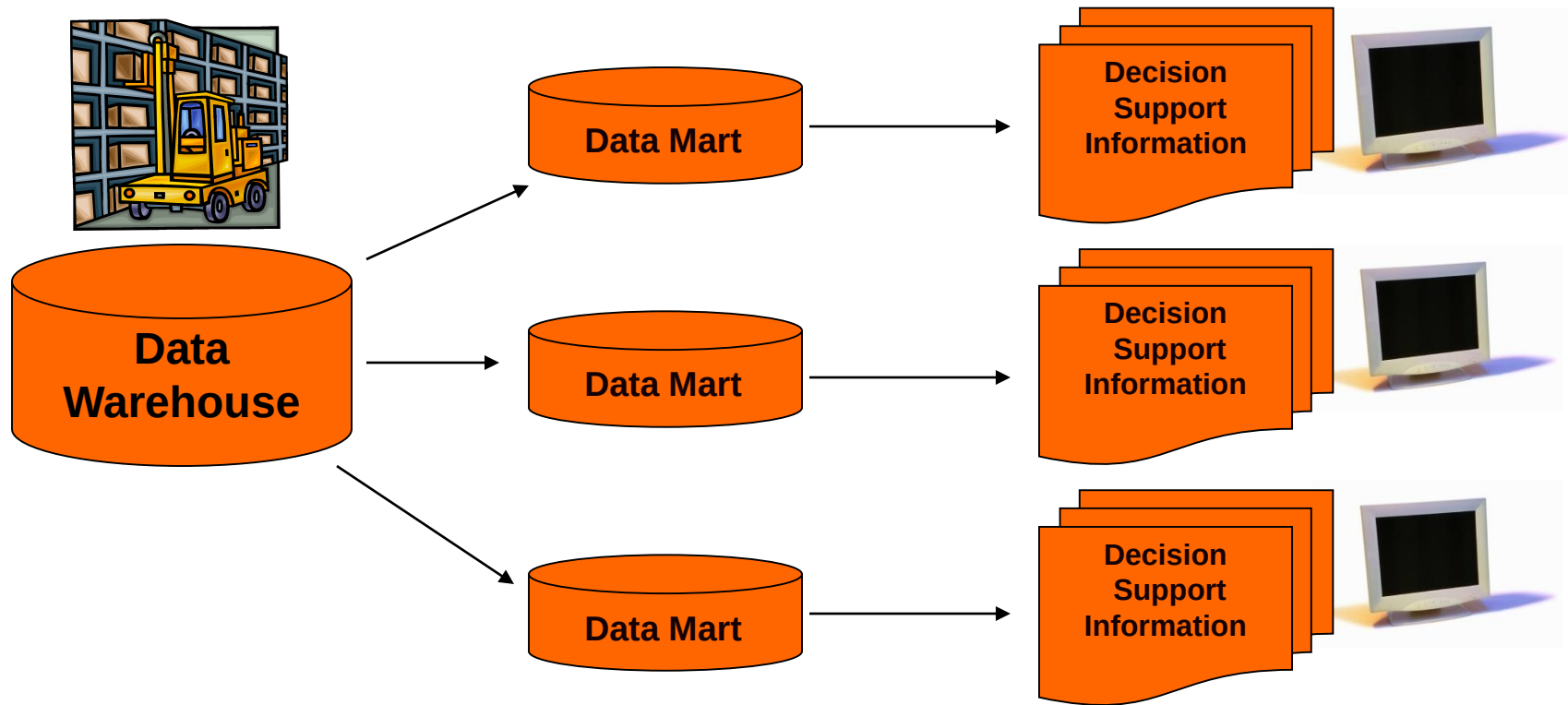


Data Mart



- ⌘ A Data Mart is a smaller, more focused Data Warehouse – a mini-warehouse.
- ⌘ A Data Mart typically reflects the business rules of a specific business unit within an enterprise.

Data Warehouse to Data Mart



Data warehouse construction



⌘ Heterogeneous schema integration

- ☒ merge from various sources
- ☒ remove inconsistencies

⌘ Data cleaning:

- ☒ missing data, clean fields e.g. names/addresses

⌘ Data loading: efficient parallel loads

⌘ Products: Prism warehouse manager, Platinum info refiner, info pump, QDB.

Warehouse maintenance



⌘ Data refresh

📁 when to refresh, what form to send updates?

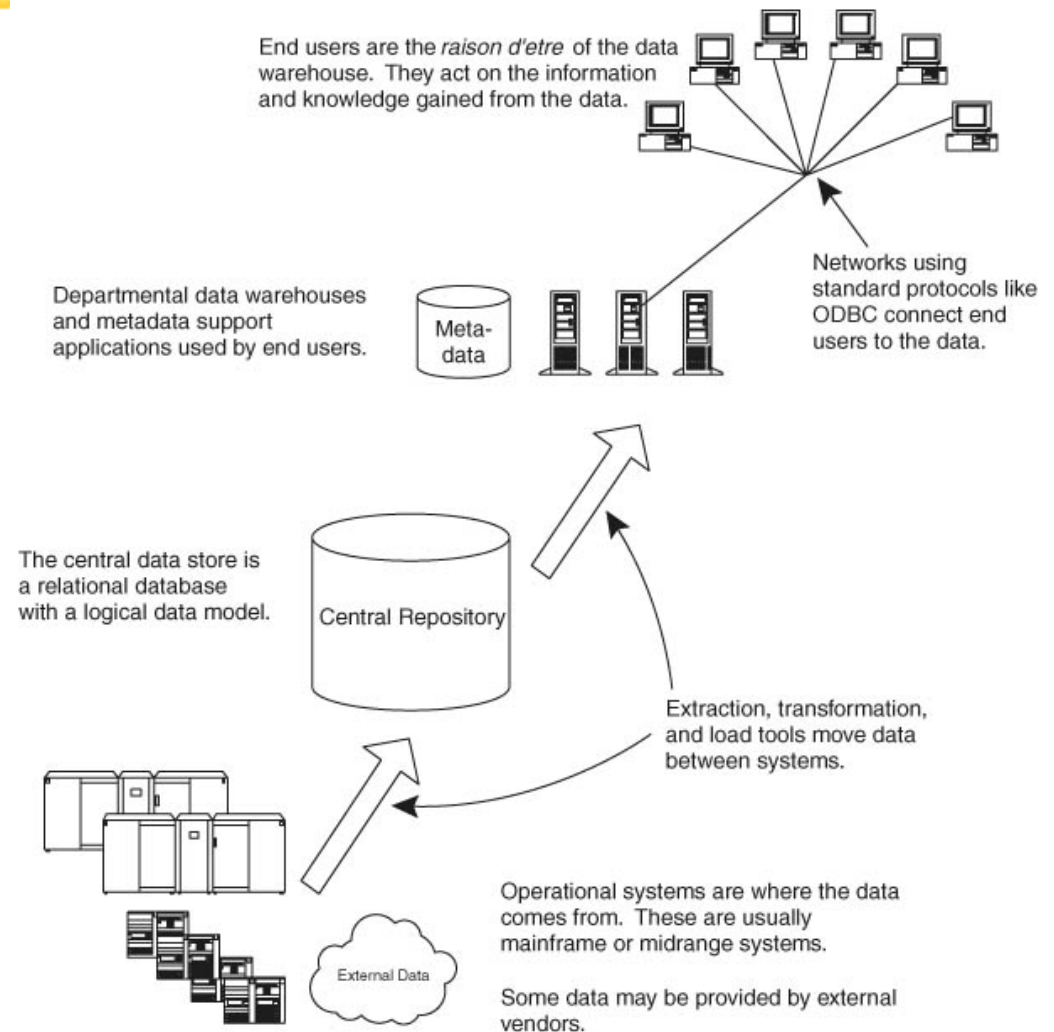
⌘ Materialized view maintenance with batch updates.

⌘ Query evaluation using materialized views

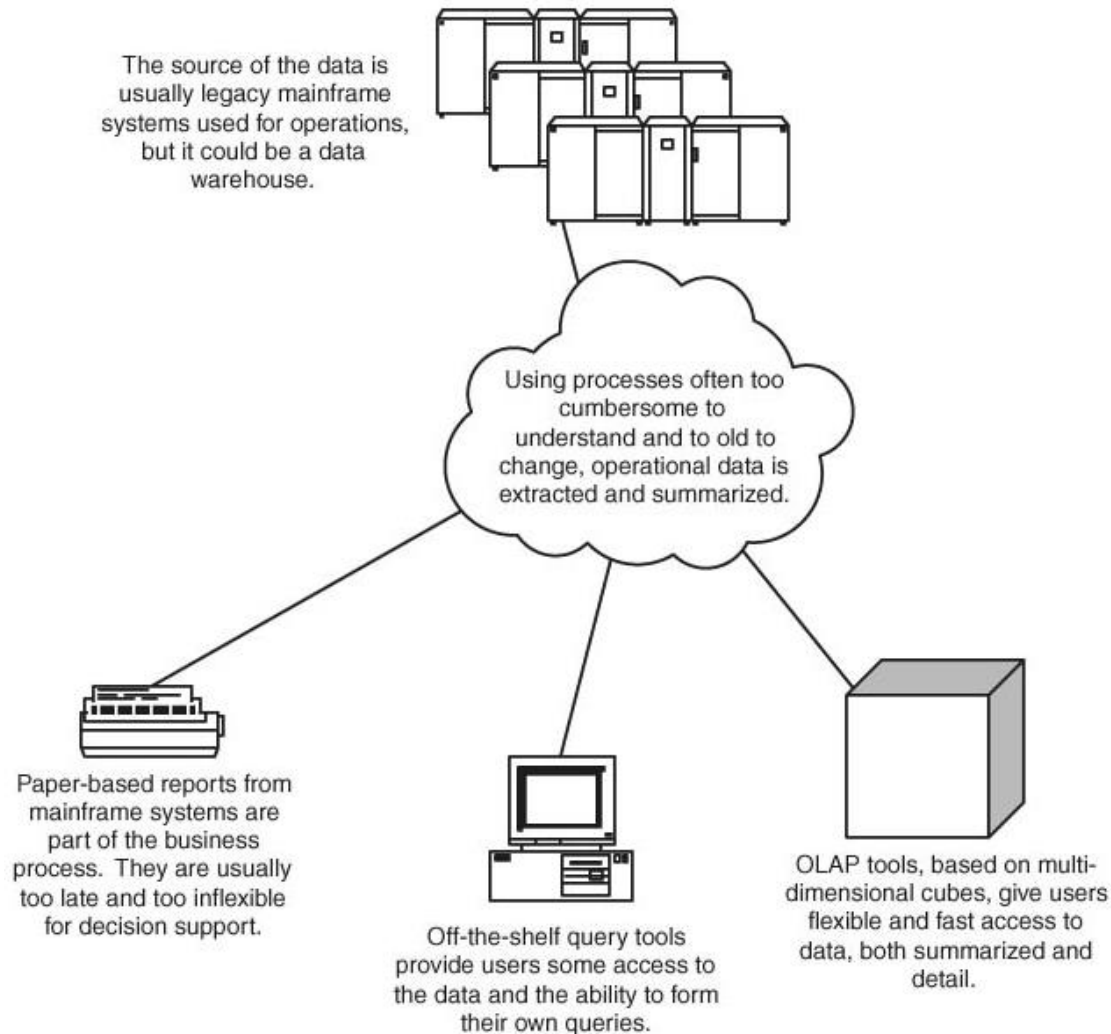
⌘ Monitoring and reporting tools

General Architecture for Data Warehousing

- ⌘ Source systems
- ⌘ Extraction, (Clean), Transformation, & Load (ETL)
- ⌘ Central repository
- ⌘ Metadata repository
- ⌘ Data marts
- ⌘ Operational feedback
- ⌘ End users (business)



Where does OLAP fit in?



OLAP



Fast, interactive answers to large aggregate queries.

⌘ Multidimensional model: *dimensions* with *hierarchies*

⌘ Dim 1: Bank location:

⌘ branch-->city-->state

⌘ Dim 2: Customer:

⌘ sub profession --> profession

⌘ Dim 3: Time:

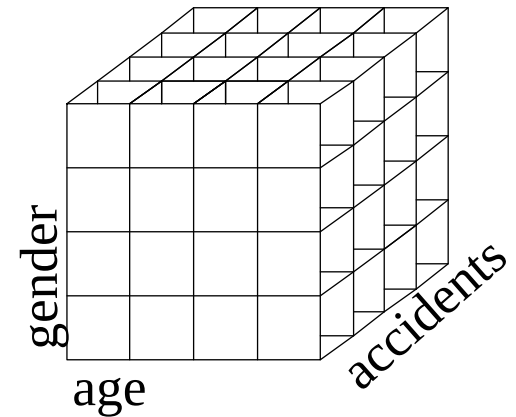
⌘ month --> quarter --> year

⌘ *Measures*: loan amount, #transactions, balance

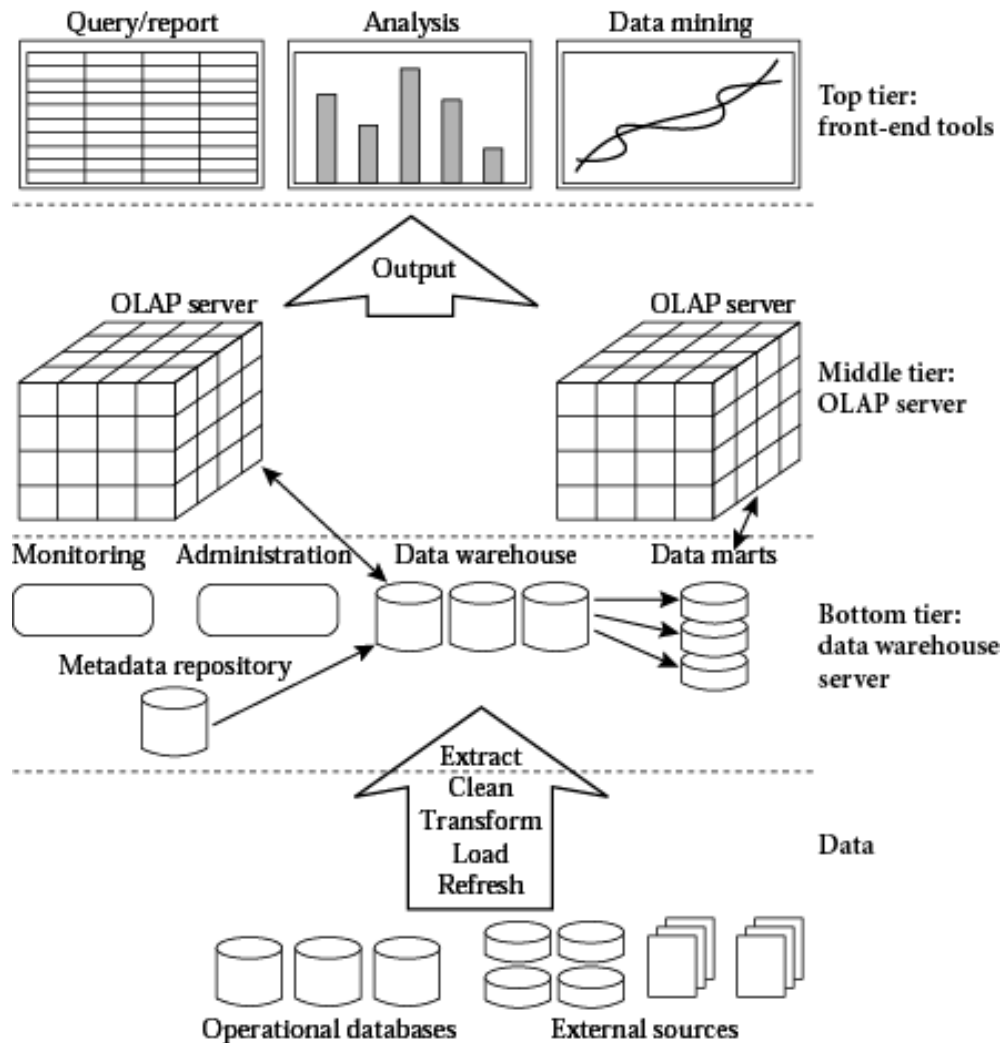
OLAP Overview

⌘ Interactive, exploratory analysis of multidimensional data to discover patterns

gender	age	accident
Male	27	3
Male	37	1
Male	37	0
Male	37	1
Male	49	2
Male	39	4
Male	43	0
Male	41	2
Male	49	1
Male	44	2
Male	43	3
Male	53	4
Male	60	0
Female	26	0
Female	39	0
Female	45	2
Female	41	2
Female	39	1
Female	37	0
Female	43	1

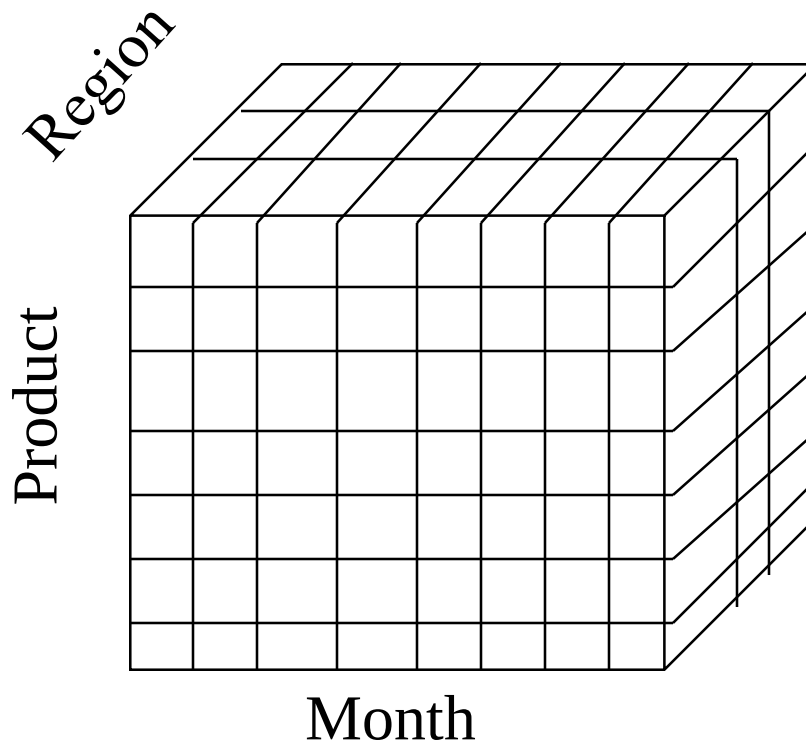


OLAP Architecture/Three-tier Data Warehousing Architecture

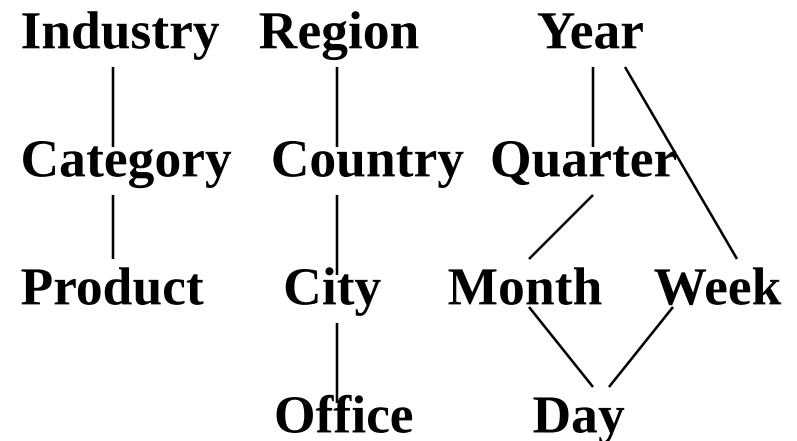


Multidimensional Data

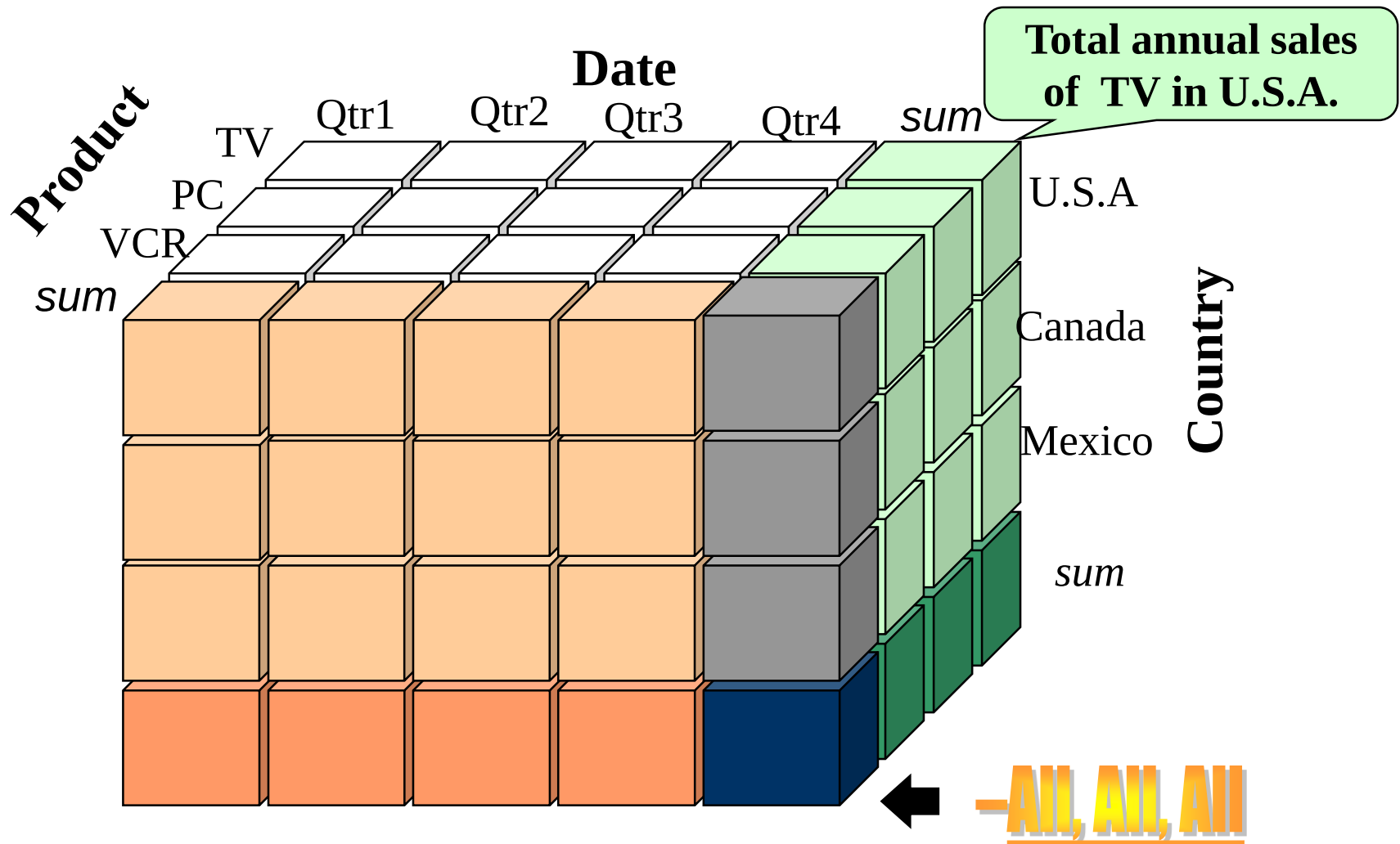
⌘ Sales volume as a function of product, month, and region



Dimensions: Product, Location, Time
Hierarchical summarization paths



A Sample Data Cube



Typical OLAP Operations

- ⌘ Roll up (drill-up): summarize data
 - ☒ *by climbing up hierarchy or by dimension reduction*
- ⌘ Drill down (roll down): reverse of roll-up
 - ☒ *from higher level summary to lower level summary or detailed data, or introducing new dimensions*
- ⌘ Slice and dice:
 - ☒ *project and select*
- ⌘ Pivot (rotate):
 - ☒ *reorient the cube, visualization, 3D to series of 2D planes.*
- ⌘ Other operations
 - ☒ *drill across: involving (across) more than one fact table*
 - ☒ *drill through: through the bottom level of the cube to its back-end relational tables (using SQL)*

Rotation (Pivot Table)

		Region			
Year	Data	Asia	Europe	North America	Grand Total
1995	Sum of Hardware	97	23	198	318
	Sum of Software	83	41	425	549
1996	Sum of Hardware	115	28	224	367
	Sum of Software	78	65	410	553
1997	Sum of Hardware	102	25	259	386
	Sum of Software	55	73	497	625
Total Sum of Hardware		314	76	681	1071
Total Sum of Software		216	179	1332	1727

		Year			
Region	Data	1995	1996	1997	Grand Total
Asia	Sum of Hardware	97	115	102	314
	Sum of Software	83	78	55	216
Europe	Sum of Hardware	23	28	25	76
	Sum of Software	41	65	73	179
North America	Sum of Hardware	198	224	259	681
	Sum of Software	425	410	497	1332
Total Sum of Hardware		318	367	386	1071
Total Sum of Software		549	553	625	1727

OLAP Server Architectures



⌘ Relational OLAP (ROLAP)

- ☒ Use relational or extended-relational DBMS to store and manage warehouse data and OLAP middle ware to support missing pieces
- ☒ Include optimization of DBMS backend, implementation of aggregation navigation logic, and additional tools and services
- ☒ greater scalability

⌘ Multidimensional OLAP (MOLAP)

- ☒ Array-based multidimensional storage engine
- ☒ fast indexing to pre-computed summarized data

⌘ Hybrid OLAP (HOLAP)

- ☒ User flexibility, e.g., low level: relational, high-level: array

⌘ Specialized SQL servers

- ☒ specialized support for SQL queries over star/snowflake schemas



Thank You